

Cholesterol and Heart Disease

Does high cholesterol cause heart disease? Learn about the relationship between high cholesterol and heart disease and how you can reverse its effects for the sake of your heart.

📄 Article last reviewed 24 May 2021

🕒 4 mins read



What Is Cholesterol?

Cholesterol is a waxy, fat-like substance found in the cell walls of our body, from the nervous system to the liver to the heart. The body uses cholesterol to make hormones, bile acids, vitamin D and other substances.

Causes of High Blood Cholesterol

Various factors may contribute to unhealthy cholesterol levels:

- Heredity
- Age

- Sex
- Diet (particularly the intake of foods rich in saturated fat, trans fat, cholesterol)
- Being overweight/obesity
- Physical inactivity

Where Does Cholesterol Come From

Our liver produces most of the cholesterol found in our body; the rest of the cholesterol is contributed by the foods we eat.

Dietary cholesterol comes only from foods of animal origin, such as liver and other organ meats, egg yolks (but not the whites, which have no cholesterol), shrimp and whole milk dairy products including butter, cream and cheese.

Types of Cholesterol

Cholesterol circulates in the bloodstream in packages called lipoproteins, which is a combination of cholesterol and protein. There are two main types of lipoproteins:

- Low-density lipoprotein (LDL), commonly known as “bad” cholesterol as it carries cholesterol to tissues via blood vessels and arteries. This “bad” cholesterol causes heart disease by building up in coronary arteries
- High-density lipoprotein (HDL), which is also called “good” cholesterol as it takes cholesterol from tissues to the liver, hence removing it from the body

Does Cholesterol Cause Heart Disease

High blood cholesterol causes heart disease when cholesterol starts building up in the coronary arteries of the heart. The relationship between cholesterol and heart disease is cumulative and gradual — if there is excess cholesterol in the blood, some of it can become trapped within the artery walls. Over time, this builds up as “plaque”, which narrows blood vessels and makes them less flexible, a condition called atherosclerosis or hardening of the arteries.

This can happen to blood vessels anywhere in the body, including those of the heart, which are called the coronary arteries. When atherosclerosis affects the coronary arteries, the condition is called coronary heart disease or coronary artery disease.

You can read more related articles on the SingHealth websites:

Cholesterol

[Can Plant Sterols Reduce LDL \(Bad\) Cholesterol? - HealthXchange.sg](#)

[High Blood Cholesterol & Hyperlipidaemia – Prevention, Treatments](#)

Read these next:

- [Coronary Artery Disease](#)
- [Heart Failure: Basic Dietary Guideline](#)



Contributed By

Singapore General Hospital